

Statement of Work (SOW) for Hypobaric Chamber Pilot Studies

The objective of this pilot hypobaric chamber testing is to support NASA decompression sickness (DCS) testing. Facilities are to perform 3 test days, collecting and recording data each day as they would for a research protocol. The first day of testing may be considered a practice session with the recognition that some facility and/or hardware integration issues may arise. The second day of testing should be scheduled after issues from the previous run have been resolved, and testing should be completed without interruption, unless otherwise indicated. The third day of testing should occur in the same work week as the second test day to demonstrate the facility's capability to perform multiple runs within one week, a capability that is needed to meet NASA testing throughput and schedule to support Artemis.

NASA's own facilities will support this work in part but are shared with other operationally relevant tasks/programs, limiting NASA facilities' overall availability and impacting scheduling and completion of critical work. Identification and validation of non-NASA resources to support these DCS research objectives are important steps to facilitating safe and successful prebreathe procedures and extravehicular activity (EVA) in weightlessness and during exploration missions.

Under this Firm Fixed Price (FFP) contract, the contractor shall provide all necessary management, personnel, training, materials, equipment, and facilities to complete all required tasks and deliverables.

Subjects

Facilities will recruit and screen applicants to provide minimum of 2 subjects for each of the 3 test days to fulfill the requirements described above.

Each test day is anticipated to last approximately 13 hours. Each of the 3 single-day runs will have unique subjects. Screening back-up subjects in case of subject availability issues, withdrawal, or disqualification is highly recommended.

Facilities should screen subjects per their standard protocols, and the subject pool should include females to demonstrate their capabilities to recruit and test this important population. NASA will provide inclusion and exclusion criteria for a population demographic similar to the astronaut corps. As part of the screening process, subjects must pass a medical examination similar to a private-pilot flight physical and complete a peak (maximal) oxygen consumption ($\text{VO}_{2\text{pk}}$) test using a cycle ergometer protocol with metabolic gas analysis. NASA will provide the $\text{VO}_{2\text{pk}}$ cycle ergometry test protocol. To qualify for participation, subjects must achieve $\text{VO}_{2\text{pk}} > 27.5 \text{ ml/kg/min}$.

Following the screening, subjects will participate in a familiarization session during which they will receive verbal and written description of the test protocol, be encouraged to ask questions, and will provide written informed consent before proceeding to participation. Thereafter, subjects will be familiarized with the chamber protocol and safety procedures, the appropriate mask size will be determined, and the subjects will practice mask breathing. Subjects also will be familiarized with the EVA simulation workstations, and metabolic rate will be measured while performing all 8 activity/rest stations (submaximal exercise intensity) to verify the cadence or work rates required for the in-chamber exercise prescription.

Standardized Test Day

Facilities will implement a test plan that includes a 6.5-hour prebreathe followed by a simulation of EVA lasting 6 hours, or until signs and symptoms that require test termination (per NASA and facility standards), test termination by subject request, or facility or monitoring capabilities malfunctions. These sessions will include successful mask operations and demonstration of safety procedures. An additional 30 minutes should be allotted for repressurization and post-test monitoring.

Medical Monitoring

The facility will provide the appropriate medical support to perform pre-chamber physical examination before ingress, during the protocol, for treatment and management of subjects as needed, clearance of subjects after the protocol is completed, and follow-up in the days after the chamber run, as required. Responsibilities of the medical personnel include monitoring and grading of 2-D ultrasound venous gas emboli as well as monitoring of the subject's self-report and symptoms of DCS or intolerance.

Prebreathe Conditions

Facilities will demonstrate the capabilities to perform 100% O₂ prebreathe durations up to 6.5 hours. Subjects will begin mask breathing at the start of the prebreathe and remain on mask through decompression and until repressurization is completed. Subjects should be provided fluids *ad libitum* without a break in prebreathe (using drink port). NASA will provide breathing masks or provide approval for a suitable substitution. Food will be allowed prior to start of mask prebreathe but not during the test protocol (for 13 hours). Subjects also are not allowed to sleep during the test protocol.

EVA Simulation

Facilities will demonstrate their capacity to perform a simulated 6-hour EVA and maintain standardized monitoring practices. To simulate the suit environment, the chamber will depress to 5.2 psia before initiating the EVA simulation tasks described below. The start time of the EVA is defined by the start of depressurization, and the end time of the EVA is defined by the start of repressurization. Target press/depress rates will be at an average 3000ft/min, not to exceed 5000ft/min

EVA simulation will include rotation of each subject through a repetitive series of tasks at predetermined intervals in a pre-determined order using NASA-defined or NASA-provided workstations (Figure 1). The EVA is simulated using eight stations, nominally performed in 2 cycles of 4 stations, with subjects rotating between stations every 5 minutes. In one cycle, subjects will rotate through stations that simulate vigorous ambulation (high step), upper body technical tasks (standing board), and load carrying (weight transfer), and 2-D ultrasound while subjects are lying left-lateral decubitus on a cot. In the second cycle, subjects will rotate through stations that simulate less vigorous ambulation (low step), upper body physical exertion (arm ergometry), technical tasks performed in a kneeling position (kneeling board), and 2-D ultrasound while subjects are lying left-lateral decubitus on a cot. Thus, in each cycle subjects will perform 15 minutes of low- to moderate-intensity exercise followed by 5 minutes of monitoring and rest. Work rates during arm ergometry (cadence and resistance) and stepping (cadence) will be prescribed based upon the results of the VO₂pk test and the pre-test EVA simulation familiarization with metabolic gas analysis. During ultrasound imaging, at least 10 cardiac cycles will be monitored and recorded in each of three conditions in sequential order: rest, flexing/bending of the legs, and rest. Compliance of the subjects with the prescribed procedures should be documented, including the time at each station, work rates (cadence and load), symptoms reported by the subjects, and test operator observations.

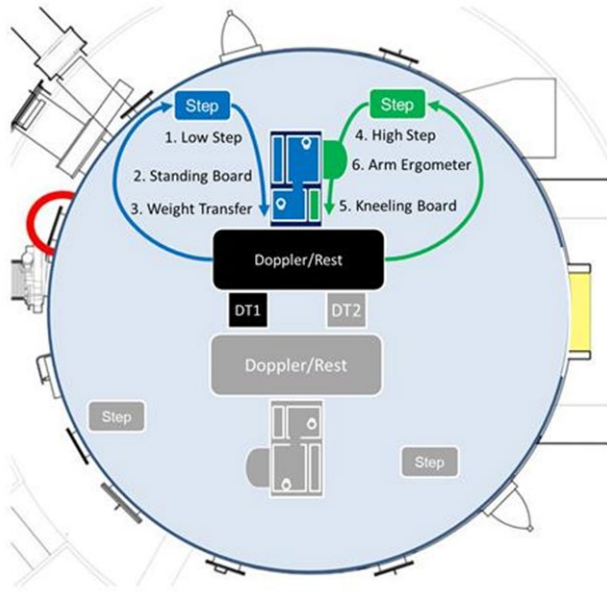


Figure 1. Example chamber layout and ordering of EVA workstations

Standardized Measures

Facilities will use a standardized set of measurements to monitor and document test subjects during the prebreathe, decompression, and recompression.

Logging of symptoms and definition of DCS

Test operators will record all symptoms reported by the test subjects throughout the duration of the protocol, whether in response to prompting by the test operators or spontaneous self-reporting. Subjects will be removed from the chamber and repressurized (airlock or chamber) at the onset of DCS (Type I or Type II).

NASA (JPR 1800.3 E) defines DCS severity with the following criteria:

- **Mild DCS (Type I), Uncomplicated:** Symptoms involving joint pain, peripheral nervous system, or simple skin bends that resolve upon repress or within the first 20 minutes of treatment.
- **Mild DCS (Type I), Complicated:** Symptoms involving joint pain, peripheral nervous system, or simple skin bends that DO NOT resolve upon repress or within the first 20 minutes of treatment.
- **Mild DCS (Type I) – Repetitive:** Mild DCS symptoms involving joint pain, peripheral nervous system, or simple skin bends that occur after a successful treatment and within 30 days of a prior case of DCS for ground-based exposure. These are considered discrete occurrences of DCS in response to chronologically discrete exposures.
- **Serious DCS (Type II):** Symptoms involving the Central Nervous System (CNS), cardiovascular system (circulatory collapse/shock), pulmonary system (chokes).

NOTE: Previous studies using similar procedures at other facilities had rates of Type I DCS between 11% and 50% when simulated extravehicular activities were conducted at 4.3 psi for 6 hours following a 6.5 hour prebreathe. All cases of DCS were resolved with ground-level oxygen treatment.

Questionnaires

Test operators will query subjects for responses to standardized questionnaires during each rotation through the resting/ultrasound station. Subjects will report a rating of perceived exertion (RPE) and

pain using NASA-defined scales at the end of each exercise bout (i.e., after 5 minutes of arm ergometry, stepping).

Cardiac Ultrasound

2-D imaging for detection of venous gas emboli will be performed by a qualified test operator or subjects trained to self-scan during each rotation through the resting/ultrasound station while the subject is at rest and following exercise. 2-D ultrasound will be recorded and stored electronically with relevant metadata (e.g., subject ID) for post-exposure review and analysis. Ultrasound will be scored using the NASA-modified Eftedal-Brubak (EB) scale (0-7) (1).

Pulse Oximetry (Recommended)

Oxygen saturation (S_pO_2) should be monitored during the decompression, EVA simulation, and recompression. Test operators must be able to observe the current S_pO_2 from the device as needed and will record S_pO_2 at frequent intervals. Continuous monitoring is not required.

Heart Rate (Recommended)

Heart rate should be continuously monitored and recorded using chest-worn devices (e.g., Polar/Garmin heart chest strap) from which data can be downloaded and stored electronically with relevant metadata (e.g., subject ID, date, time) for post-exposure analysis. Test operators must be able to observe the current heart rate from the device as needed and will manually record heart rate at the end of each 5-minute rotation through the EVA simulation stations.

Data Validation

At the conclusion of each test day, the facility will deliver electronic records to NASA using a secure, NASA-approved method. The data package will include subject metadata, console and/or operator's logs (including chamber event time, prebreathe records, EVA time, EVA station schedule), chamber pressure recordings (demonstrating depress and repress rates as well as sustained chamber pressure), chamber environmental conditions (partial pressures of O_2 , CO_2 , and N_2 , temperature, humidity), questionnaires, medical monitoring logs, heart rate and pulse oximetry records (if recorded), Doppler and ultrasound recordings, in-chamber video of EVA workstations, and DCS case descriptions (including DCS symptom description/severity, DCS onset timing, DCS treatment timing/pressure, and DCS resolution timing/pressure).

Each facility will have ultrasound images scored by 3 independent reviewers (i.e., not aware of the other reviewers' scoring) and blinded to the subject and time/pressure profile. NASA can provide a template for the documentation of these.

Test Plan Verification, Training, and Reporting

NASA will meet with each facility to discuss the test plan before submission of required documents to facility's governing IRB (and NASA, as required) to verify expectations of the chamber pilot runs (Figure 2) and the notional schedule (Table 1). NASA will provide limited real-time technical beyond that support for this work beyond that described in this section. Suppliers should include costs for a virtual technical interchange meeting (TIM) for final data coordination and summary of lessons learned.

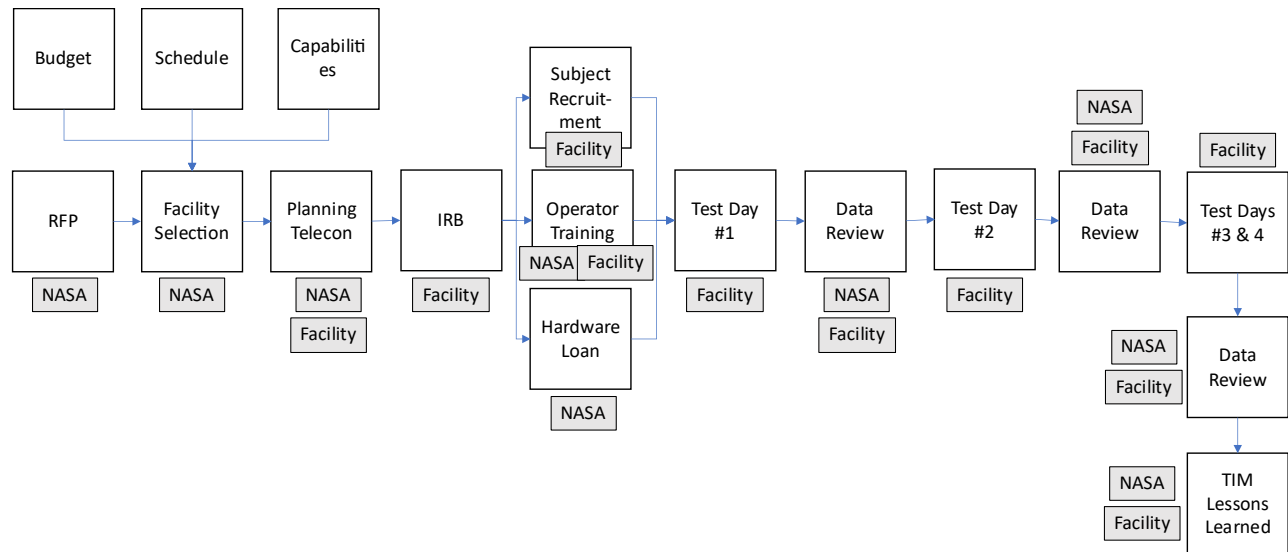


Figure 2. Schematic of the demonstration project. Shaded boxes show entities primarily responsible for each activity.

Table 1. Notional Schedule

Milestones	FY26		FY27			
	Q3	Q4	Q1	Q2	Q3	Q4
Experiment Prep (IRB, coordination with NASA, etc.)						
Facility procures/constructs EVA Task elements						
NASA Delivers Loaner Masks and Equipment (if required)						
Chamber Runs and Resolution of Deficiencies						
Data Delivery and Technical Close-out Meeting						

Note: Providers will coordinate the actual schedule for this work with NASA. Schedule above is only notional. Plans will be made to loan hardware (e.g., breathing masks, ultrasound), if needed, and scheduling for operator training.

NASA plans to have capability to loan the following equipment as Government Furnished Equipment GFE.

- Ultrasound (GE Vivid IQ)
- Masks (EATON POM Plus)

For each facility, NASA also will provide training in:

- Mask fitting procedures,
- EVA task simulation, and
- Operation of loaner equipment (if required).

NASA will monitor at least one chamber run, preferably one of the first two, to answer any questions in real-time and recommend any changes to the procedures and may also monitor the final run (ie that

demonstrates all goals achieved). Travel/lodging of NASA personnel will be handled by NASA; but the facility shall handle necessary site access procedures. At the completion of the chamber runs, NASA will coordinate a meeting with each facility to discuss results of these pilot runs and forward plans.

Work described above must be completed within 5 months of Institutional Review Board (IRB) approval.

Optional Work:

Vendors will provide optional work costs for the following efforts:

Detailed protocols for optional efforts will be provided by NASA after the base study (assume measures match base effort).

CLIN 0002:

A fourth test day with minimum of two subjects.

CLIN 0003:

Break in Prebreathe with “inflight-like” treatment Preparation

-An additional three days of testing, with each test day simulating DCS treatment capabilities. This will include simulating a subject with DCS and providing either the inflight-like treatment and/or the gold-standard treatment. The time to treatment should be logged for each day.

CLIN 0004:

Break in Prebreathe with “inflight-like” treatment

-Using the protocol described above, data collection and delivery for an additional 50 subjects, each participant completes testing with a break in prebreathe protocol. Any subjects with DCS shall receive inflight-like treatment at 8.2 psia (initially with for a designated period of time followed by 16.4-17.8psi (exact pressures and durations to be provided by NASA), with options to proceed to standard of care (i.e., ground level oxygen or hyperbaric oxygen therapy) at any time during the test or inflight like treatment.

CLIN 0005:

Elevated Metabolic Rate during EVA Testing: Using the protocol described above, data collection and delivery for an additional 24 subjects, each participant completing the 6-hour simulated EVA on three different days at three different work rates each (i.e., different average EVA metabolic rate) days in a cross-over study design. The proposer should assume that the same equipment will be used for the EVA but subjects will perform EVA tasks at a different pace on each test day to result in a different average metabolic rate. NASA will provide the details of the EVA work rate prescriptions for the three test days.

CLIN 0006:

Pharmaceutical-enhanced Prebreathe: Using the protocol described above, data collection and delivery for an additional 24 subjects, each participant completing testing with three different pharmaceutical countermeasures on three different test days in a cross-over study design. NASA will provide the pharmaceutical and the details for their administration.

CLIN 0007:

Body Temperature: Using the protocol described above, data collection and delivery for an additional 24 subjects, each participant completing testing with three different temperatures as a countermeasure for DCS on three different test days in a cross-over study design. NASA will provide the details for the temperature profile.

REFERENCES

1. **Eftedal O, Brubakk AO.** Agreement between trained and untrained observers in grading intravascular bubble signals in ultrasonic images. *Undersea Hyperb Med* 24: 293–299, 1997.